

FNCE 5352 — Lecture 4: Linear Regression (Part 2)

Matthew McDonald

Where Are We?

We now have working examples of:

- Simple regression (Advertising, AAPL ~ SPY)
- Multiple regression (+ predictors, "holding other vars constant")
- Performance metrics (RMSE / MSE / R^2), incl. [yardstick](#)

Important questions (ISLR Ch. 3)

The “important questions” checklist

Even when the output looks clean:

- Is the relationship linear?
- Are residuals “well-behaved”?
- Are there outliers or high-leverage points?
- Are predictors correlated (collinearity)?
- Are we evaluating fairly (no peeking)?

Linear model assumptions (in English)

A linear model is most useful when:

- the mean relationship is roughly linear
- residuals don't have obvious structure
- noise is roughly stable across the range (often false in finance)
- individual observations aren't dominating the fit

Residual plots: what we look for

Residuals vs fitted values:

- random scatter around 0 (good)
- curve pattern (missing nonlinearity)
- funnel / changing spread (heteroskedasticity)
- a few huge residuals (outliers / events)

Finance translation

If residuals show patterns, possible stories:

- regime shifts / volatility clustering
- missing risk factor exposure
- nonlinear exposure (beta changes in stress)
- event days (earnings, macro surprises)

Don't tune on the test set

- Keep a test set you look at *once* (or rarely)
- Use train/validation (or resampling) for model choices
- In finance: **time split** → the future is the test

Performance metrics: what they are (and what they aren't)

- **RMSE**: typical size of prediction error (units of Y)
- **MSE**: same, but squared units (penalizes large errors)
- **R²**: variance explained (always specify in-sample vs out-of-sample)

Yardstick as a metrics calculator

Why `yardstick` is handy (even using base `lm()`):

- consistent definitions (no copy/paste metric bugs)
- clean “truth vs estimate” interface
- works nicely with tibbles



Estimating Performance

The **yardstick** package is a tidy interface for computing measures of performance.

There are individual functions for specific metrics (e.g. **accuracy()**, **rmse()**, etc.).

When more than one metric is desired, **metric_set()** can create a new function that wraps them.

Note that these metric functions work with **group_by()**.

```
# yardstick loaded by tidymodels

perf_metrics <- metric_set(rmse, rsq, ccc)

# A tidy result back:
test_pred %>%
  perf_metrics(truth = log_price, estimate = .pred)
```

```
## # A tibble: 3 x 3
##   .metric .estimator .estimate
##   <chr>    <chr>      <dbl>
## 1 rmse     standard      0.155
## 2 rsq      standard      0.181
## 3 ccc      standard      0.306
```

There are sometimes different ways to estimate these statistics; **.estimator** is not always "standard".

Other considerations

Correlated predictors (collinearity)

When X's move together:

- coefficients can swing a lot with small data changes
- “holding other vars constant” becomes fragile
- prediction can still be OK even if interpretation is unstable

Outliers and leverage (conceptual)

- an outlier in Y: big residual (may not move the line much)
- a high-leverage point: unusual X (can move the line a lot)
- finance: single days can dominate if you let them

Qualitative predictors (categorical X)

- `lm()` turns factors into indicator (dummy) variables
- coefficients are differences vs a baseline level
- interpretation: "difference holding other vars fixed"

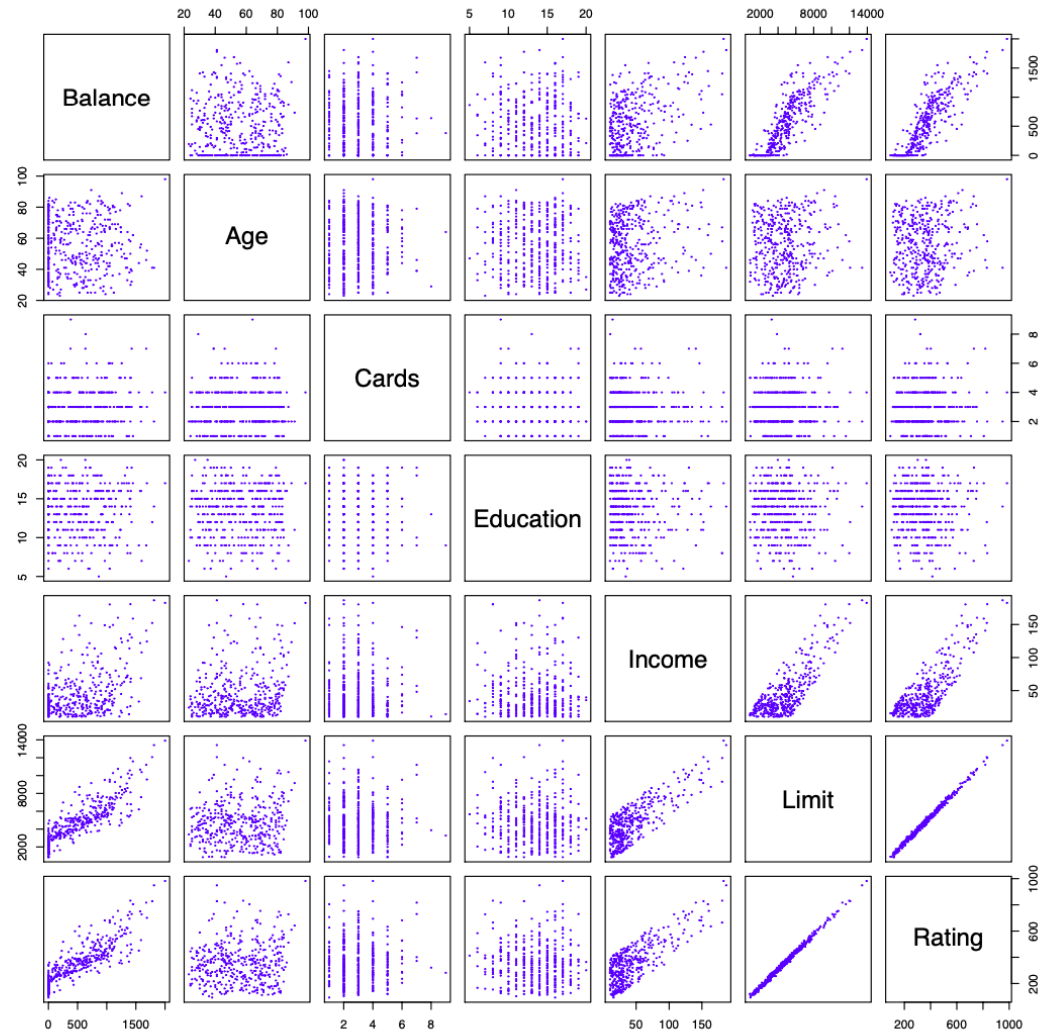
Other Considerations in the Regression Model

Qualitative Predictors

- Some predictors are not *quantitative* but are *qualitative*, taking a discrete set of values.
- These are also called *categorical* predictors or *factor variables*.
- See for example the scatterplot matrix of the credit card data in the next slide.

In addition to the 7 quantitative variables shown, there are four qualitative variables: **gender**, **student** (student status), **status** (marital status), and **ethnicity** (Caucasian, African American (AA) or Asian).

Credit Card Data



Interactions and nonlinearity

- interactions: the effect depends on another variable ($X_1 \times X_2$)
- nonlinearity: transforms or richer models
- we'll revisit when we build features intentionally

What we did not cover

Outliers

Non-constant variance of error terms

High leverage points

Collinearity

See text Section 3.33

Generalizations of the Linear Model

In much of the rest of this course, we discuss methods that expand the scope of linear models and how they are fit:

- *Classification problems:* logistic regression, support vector machines
- *Non-linearity:* kernel smoothing, splines and generalized additive models; nearest neighbor methods.
- *Interactions:* Tree-based methods, bagging, random forests and boosting (these also capture non-linearities)
- *Regularized fitting:* Ridge regression and lasso